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Developmental Biology

Genetic Inactivation of the β1 adrenergic receptor prevents Cerebral Cavernous Malformations in zebrafish

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eLife Assessment

In this **important** study, the authors test the model that a type of vascular lesion caused by the inactivation of one gene in the cells that line blood vessels requires the activity of a second gene for the lesions to form. The evidence supporting the conclusions is **solid**.

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Abstract

Propranolol reduces experimental murine cerebral cavernous malformations (CCMs) and prevents embryonic caudal venous plexus (CVP) lesions in zebrafish that follow mosaic inactivation of *ccm2*. Because morpholino silencing of the β1 adrenergic receptor (*adrb1*) prevents the embryonic CVP lesion, we proposed that *adrb1* plays a role in CCM pathogenesis. Here we report that *adrb1*$^{-/-}$ zebrafish exhibited 86% fewer CVP lesions and 87% reduction of CCM lesion volume relative to wild type brood mates at 2dpf and 8-10 weeks stage, respectively. Treatment with metoprolol, a β1 selective antagonist, yielded a similar reduction in CCM lesion volume. *Adrb1*$^{-/-}$ zebrafish embryos exhibited reduced heart rate and contractility and reduced CVP blood flow. Similarly, slowing the heart and eliminating the blood flow in CVP by administration of 2,3-BDM suppressed the CVP lesion. In sum, our findings provide genetic and pharmacological evidence that the therapeutic effect of propranolol on CCM is achieved through β1 receptor antagonism.

Introduction

Cerebral cavernous malformations (CCMs), accounting for 5-15% of cerebrovascular abnormalities, are characterized by blood-filled endothelial-lined cavities, and can cause seizures, headaches, neurological deficits, and recurrent stroke risk[1]. Familial CCMs are due to heterozygous loss of function mutations in the *KRIT1*, *CCM2*, or *PDCD10* genes with a second mutation inactivating the normal allele in random brain endothelial cells[2]. We previously developed a zebrafish CCM model using CRISPR-Cas9 to inactivate *ccm2* in a manner that replicates the mosaic genetic background of the human disease[3]. This zebrafish model exhibits two phenotypic phases: lethal embryonic caudal venous plexus (CVP) cavernomas at 2 days post-fertilization (dpf) in ~30% of embryos and histologically-typical CNS CCMs in ~100% of surviving 8 week old fish[3]. Both phases of this model, like their murine counterpart, depend on Krüppel-like factor 2 (KLF2), confirming shared transcriptional pathways. Furthermore, both phases of the zebrafish model exhibit similar pharmacological sensitivities (Supplementary Table 1) to murine and human lesions. Thus, this zebrafish model offers a powerful tool for genetic and pharmacological analysis of the mechanisms of CCM formation.

Anecdotal case reports[4–6] and a recent phase 2 clinical trial have suggested that propranolol a non-selective β adrenergic receptor antagonist benefit patients with symptomatic CCMs[7]. In previous studies, we observed this effect of propranolol in both zebrafish and murine models of CCM[8]. Importantly, propranolol is a racemic mixture of R and S enantiomers and elegant work from the Bischoff lab has implicated the R enantiomer, which lacks β adrenergic antagonism, as the component that inhibits SOX18 thereby suppressing infantile hemangiomas[9]. We previously found that the anti-adrenergic S enantiomer rather than the R enantiomer inhibited development of embryonic CVP cavernomas in *ccm2* CRISPR zebrafish[8]. Furthermore, morpholino silencing of the gene that encodes the β1 (*adrb1)* but not β2 *(adrb2)* receptor also prevents CVP cavernomas[8], suggesting that the β1 adrenergic receptor (β1AR), which primarily impacts hemodynamics[10], contributes to the pathogenesis of CCM. Here we have inactivated the *adrb1* gene to eliminate the β1AR and observed the expected reduction of heart rate and contractility which resulted in reduced blood flow through the CVP. These *adrb1*$^{-/-}$ zebrafish were protected from embryonic CVP cavernomas and, importantly, also exhibited much reduced number and volume of adult brain CCM. Furthermore, a β1-selective antagonist, metoprolol, also inhibited adult CCM suggesting that β1AR-specific antagonists may be useful for CCM treatment and will carry less potential for β2 AR-related side effects such as bronchospasm[11].

Results

β1AR is important in for development of CVP cavernomas

We previously reported that morpholino silencing of *adrb1* rescued CVP cavernomas in zebrafish embryos[8]. Because of morpholinos potential for off target effects[12, 13], we sought to confirm the β1AR's potential involvement in CCM pathogenesis by inactivating *adrb1.* We used CRISPR-Cas9 to generate an 8-bp deletion resulting in a premature stop codon at 57 bp (**Figure 1A**). To exclude potential off-target effects, the top 20 potential off-target sites predicted by Cas-OFFinder (*www.rgenome.net/cas-offinder*) (Supplementary Table 2) were sequenced and were not mutated (data not shown). We intercrossed *adrb1*$^{+/-}$ F1 offspring resulting in *adrb1*$^{-/-}$ zebrafish embryos that displayed reduced cardiac contractility (Supplementary Video 1) and decreased heart rate (Supplementary Figure 1A). Similar to *Adrb1*$^{-/-}$ mice[14], *adrb1*$^{-/-}$ zebrafish exhibited a blunted chronotropic response to a β1AR agonist, isoprenaline hydrochloride (**Figure 1B**). *Adrb1*$^{-/-}$ embryos had no obvious defects in vascular development and they survived to adulthood and were fertile.

We intercrossed *adrb*$^{+/-}$ individuals and injected one-cell stage embryos with *ccm2* CRISPR and blindly scored the presence of CVP cavernomas at 48hpf. As expected we observed CVP cavernomas in 28% of *adrb1*$^{+/+}$ embryos. In sharp contrast only 3% of *adrb1*$^{-/-}$ embryos exhibited cavernomas (**Fig. 1C and D**) indicating that loss of β1AR prevents CVP cavernomas. these observations demonstrate that the β1AR is required for embryonic CVP cavernoma formation.

β1AR mediates formation of adult CCM in the brain

Adult *ccm2* CRISPR zebrafish display highly penetrant CCMs throughout the central nervous system[3]. Brains from adult *ccm2* CRISPR fish on *adrb1*$^{-/-}$ (12 brains) or wild type (13 brains) background were treated with CUBIC (clear, unobstructed brain/body imaging cocktails and

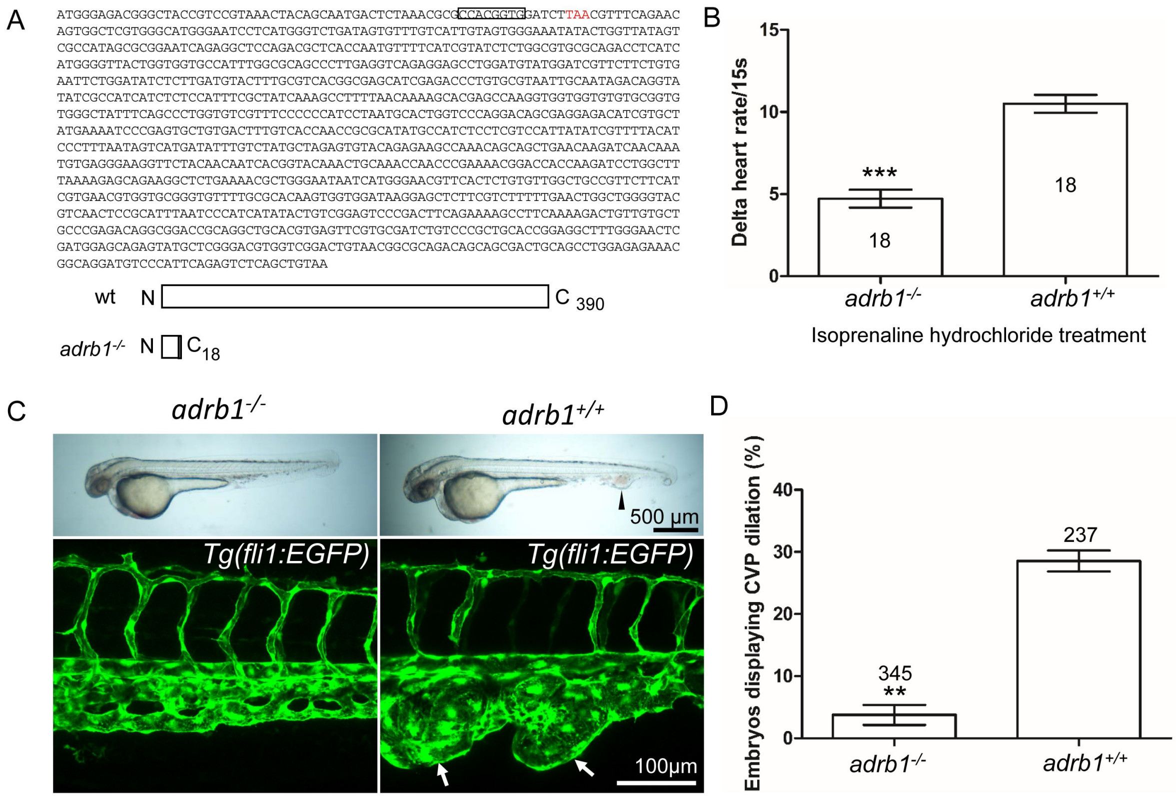

Figure 1.

Adrb1 signaling is essential for CVP dilation.

(A) The targeted *adrb1* allele shows an 8-nucleotide deletion producing a pre-stop codon. *Adrb1* null cDNA is predicted to encode truncated adrb1 protein. The wild type adrb1 protein contains 390 amino acids, while the predicted adrb1 null protein would contain 2 missense amino acids (gray bar) and would terminate after amino acid 18. (B) Isoprenaline hydrochloride (50 μM) treatment at 72hpf lead to a heart rate increase in zebrafish, while the delta heart rate in *adrb1*$^{-/-}$ is significantly smaller than that of wild type. Heartbeat was counted in 18 embryos of each group before and immediately after adding the chemical. Paired two-tailed t test, P<0.0001. (C) After *ccm2* CRISPR injection, representative bright field and confocal images of 2dpf *Tg(fli1:EGFP)* embryos show that wild type embryos display CVP dilation, while *adrb1*$^{-/-}$ embryos were resistant to this defect. Arrowhead and arrows indicate the dilation in CVP. Scale bar: 500 μm (bright field), 100μm (confocal). (D) Paired two-tailed t test shows that percentage of embryos displaying CVP dilation is significantly smaller on *adrb1*$^{-/-}$ background than that of control embryos. P=0.0012. 345 *adrb1*$^{-/-}$ embryos and 237 control embryos from 4 experiments were examined for CVP cavernoma.

computational analysis)[15], and these transparent brains were then scanned with light-sheet microscopy and lesions were enumerated and volumes were estimated with NIH ImageJ. While the typical multi-cavern CCMs were observed in brains on wild type background appearing as blood filled dilated vessels (**Figure 2A** through C), *ccm2* CRISPR *adrb1*$^{-/-}$ fish exhibited an 87% reduction in lesion volume (**Figure 2D** through G). Thus, genetic inactivation of β1AR prevented CCMs.

A selective β1AR antagonist prevents CCMs

The non-selective β blocker propranolol reduces lesion volume in murine CCM models[8,16]. To ascertain whether propranolol had a similar effect in the zebrafish model, the chemical treatment was started from larval stage at concentration which allows the fish to develop to two months for CCM inspection. We added 12.5μM propranolol to or vehicle control to fish water of *ccm2* CRISPR zebrafish starting at 3 weeks of age. The water was refreshed daily with drug or vehicle until fish were sacrificed and brains were examined as described (**Figure 3A**). Quantification based on light-sheet scanning of zebrafish brains (**Figure 3B**) showed that compared to vehicle-treated controls (**Figure 3C, D**, and **E**), propranolol-treated groups displayed a 94% reduction in CCM lesion volume (**Figure 3F, G**, and **H**). Similarly, administration of 50μM racemic metoprolol a β1-selctive antagonist produced a similar (98%) reduction in lesion volume (**Figure 3B, I, J**, and **K**). Importantly, neither drug at the doses administered reduced the growth of the fish or the volume of their brains. In sum, both genetic and pharmacological loss of β1 adrenergic receptor signaling markedly reduces the lesion burden in the zebrafish *ccm2* CRISPR model of CCM.

Loss of β1AR does not prevent increased *klf2a* expression in *ccm2* null embryos

Inactivation of CCM genes leads to increased endothelial KLF2 expression[17, 18], a transcription factor important in cavernoma formation[3, 19]. Nevertheless, silencing of *adrb1* did not prevent the expected increased endothelial *klf2a*[17] expression in *ccm2* morphant *Tg(klf2a:H2b-EGFP)* fish in which the nuclear EGFP expression is driven by the *klf2a* promoter (**Figure 4A, B** and **C**). We previously reported that mosaic expression of KLF2a occurs in *tnnt* morphant 2 dpf *ccm*2 CRISPR embryos, as judged by a widely variable in endothelial *klf2a* reporter expression[3]. Nevertheless, combination of *tnnt2a* morphant with the *adrb1* morphant (**Figure 4D**) or control morphant (**Figure 4E**) *ccm2* CRISPR *Tg(klf2a:H2b-EGFP)* embryos both displayed a similar widely variable *klf2a* reporter intensity (**Figure 4F**) indicative of similar mosaicism. Similarly, merely slowing the heart and reducing contractility with 2,3-butanedione monoxime (BDM)[20] also prevented CVP cavernomas (**Fig 4G**, and Supplementary Figure 2). Thus, silencing β1AR does not prevent the generalized increase in endothelial KLF2a in *ccm2* morphants nor does it prevent the mosaic KLF2a increase in *ccm2* CRISPR zebrafish.

Discussion

In this study, we provide genetic and pharmacological evidence to implicate the β1 adrenergic receptor (β1AR) in the pathogenesis of CCM. Our data suggest that inactivating β1AR via gene inactivation or pharmacological inhibition can significantly reduce the volume of CCM lesions in a zebrafish model.

Case reports[4–6, 21] and our previous study[8] revealed the potential benefit of the non-selective β-adrenergic receptor blocker propranolol in reducing CCMs in patients and mouse models, respectively. However, propranolol is a racemic mixture, and its R enantiomer which lacks β-adrenergic antagonism was reported to show therapeutic effect for infantile hemangiomas in an animal study[9]. Notably, our study demonstrates that *adrb1*$^{-/-}$ zebrafish are significantly

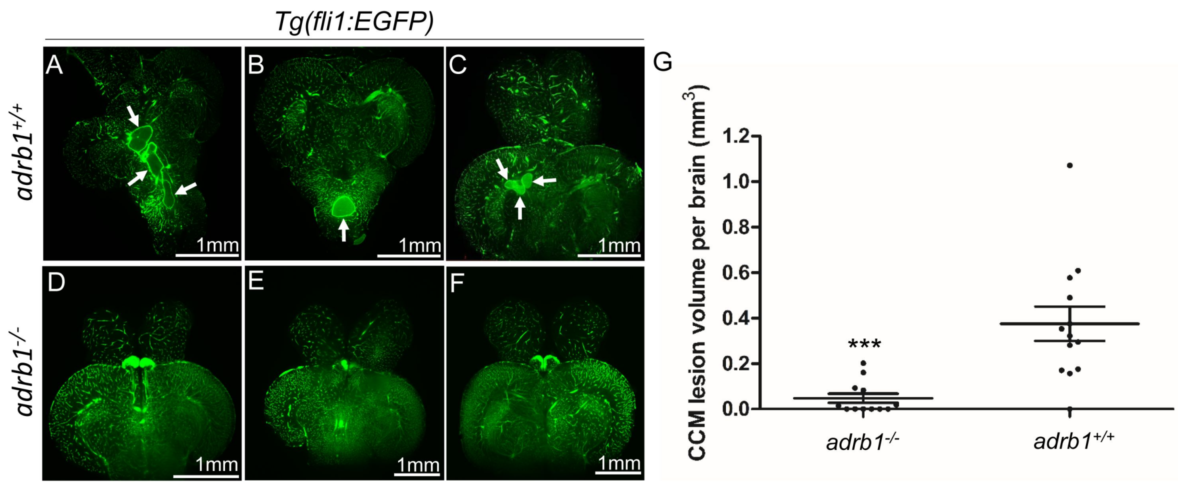

Figure 2.

Genetic inhibition of adrb1 signaling could rescue CCM in *ccm2* CRISPR zebrafish.

(A through F) Representative light sheet microscopy scanning pictures of brains from *ccm2* CRISPR adult zebrafish of *adrb1*$^{+/+}$ (A through C) and of *adrb1*$^{-/-}$ (D through F) on *Tg(fli1:EGFP)* background. Brains from *ccm2* CRISPR on wild type background show lesions indicated by arrows (A through C), while brains from *ccm2* CRISPR on *adrb1*$^{-/-}$ do not show lesions (D through F). Scale bar: 1mm. (G) Statistical analysis of total lesion volume by unpaired two-tailed t test. P=0.0005. 12 *adrb1*$^{-/-}$ brains and 13 *adrb1*$^{+/+}$ brains were analyzed.

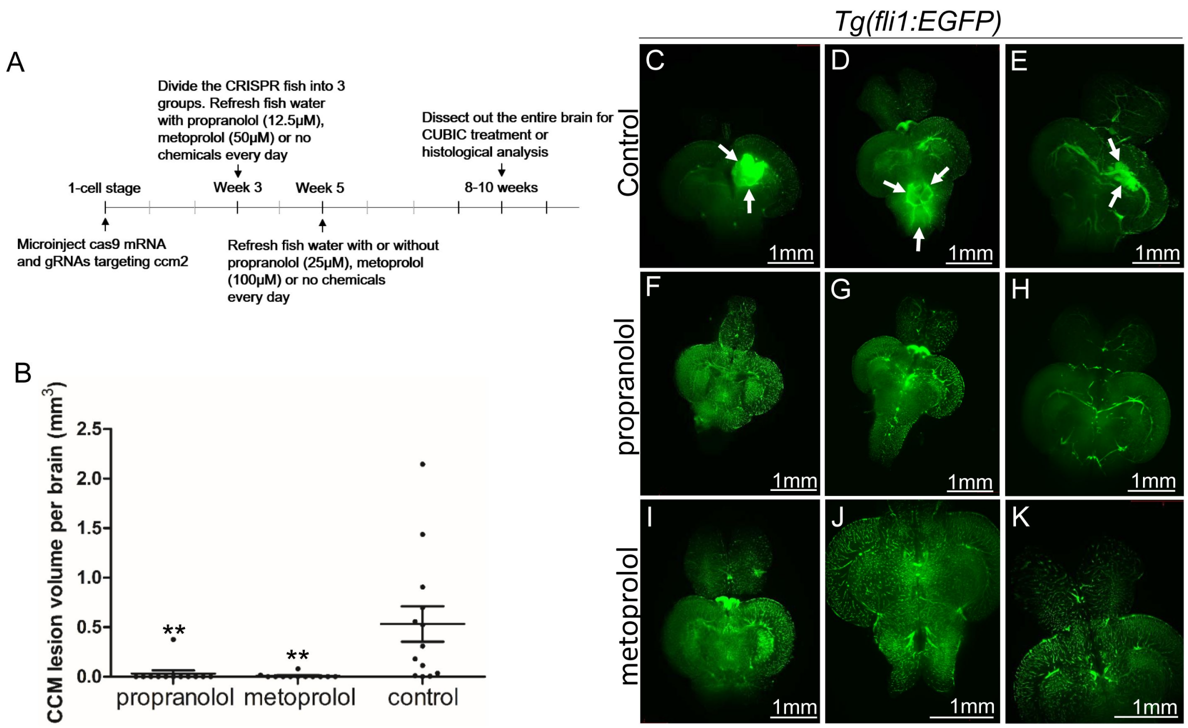

Figure 3.

Both propranolol and metoprolol could rescue CCM in *ccm2* CRISPR zebrafish.

(A) A diagram outlines the drug treatment experiment, CUBIC treatment and following recording of CCMs in adult zebrafish brain. The chemical treatment was started from week 3 with 12.5 μM propranolol or 50 μM metoprolol, and increased to 25 μM propranolol and 100 μM metoprolol, respectively from week 5. The fish water with chemicals or vehicle control are refreshed on a daily basis. (B) Statistical analysis of lesion volume by one-way ANOVA followed by Tukey's multiple comparison test. P<0.01. 12 propranolol treated, 12 metoprolol treated, and 13 vehicle brains were analyzed. (C through K) Representative light sheet microscopy scanning pictures of brains from *ccm2* CRISPR adult zebrafish on *Tg(fli1:EGFP)* background. In controls without chemical treatment (C, D, and E) there were vascular anomalies indicated by arrows. Neither propranolol (F, G, and H) nor metoprolol (I, J, and K) treated fish showed vascular lesions in the brain. Scale bar: 1mm.

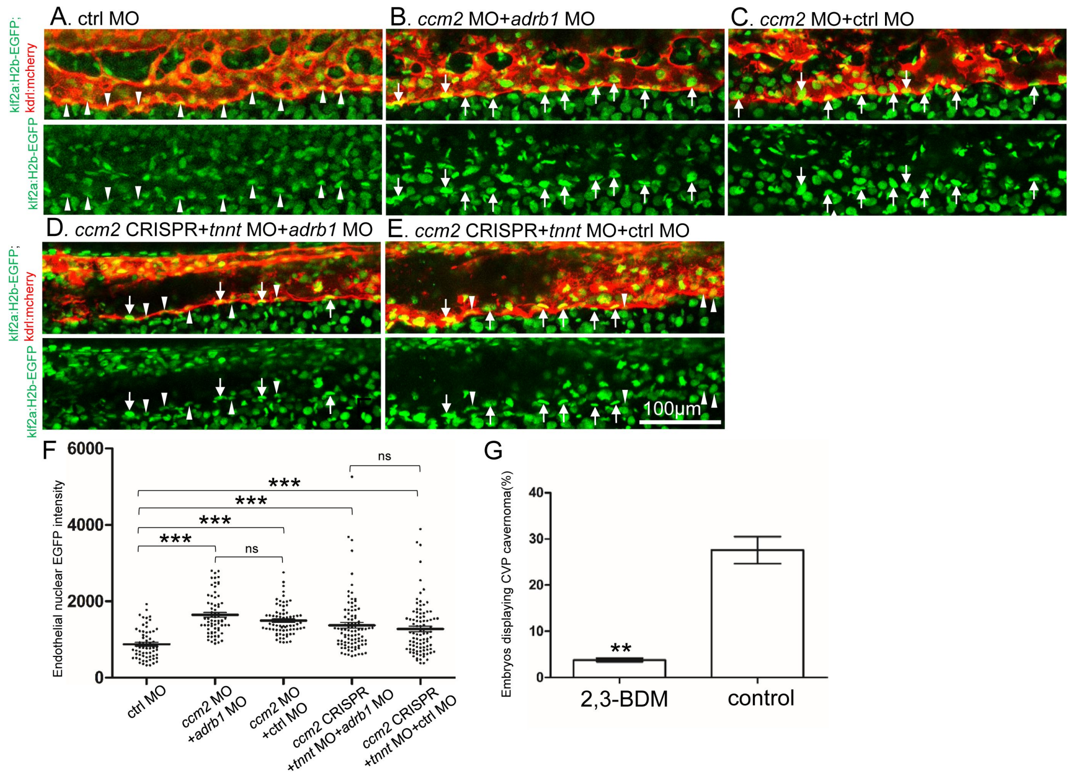

Figure 4.

Adrb1 signaling does not alter *klf2a* expression in *ccm2* CRISPR embryos.

Tg(klf2a:H2b-EGFP; kdrl:mcherry) embryos were injected and nuclear EGFP signal in mcherry labeled vascular endothelial cells is recorded by confocal. Representative images from each group are shown. (A) Control MO alone injected embryos were used as control. (B and C) *Ccm2* morphant embryos co-injected with *adrb1* MO (B) or control MO (C) both displayed significant increase of endothelial nuclear EGFP intensity (P<0.0001) compared to that of control (A), and there is no significant difference between them. (D and E) All the *ccm2* CRISPR embryos were co-injected with *tnnt* MO, which are absent of blood flow. Compared to that of control (A), *ccm2* CRISPR embryos co-injected with *adrb1* MO (D) or control MO (E) both displayed a mosaic increase of nuclear EGFP intensity of vascular endothelial cells compared to control (A) (P<0.0001), and there is no significant difference between them. Arrows indicated the endothelial nuclei with significant higher EGFP intensity than those indicated by arrowheads. Scale bar:100μm. (F) EGFP intensity of endothelial nuclei were quantified with Image J. The number of analyzed nuclei were: 63 from 10 embryos (control MO), 70 from 10 embryos (*ccm2* MO+*adrb1* MO), 77 from 10 embryos (*ccm2* MO+control MO), 93 from 13 embryos (*ccm2* CRISPR+*adrb1* MO), and 94 from 13 embryos (*ccm2* CRISPR+control MO). Statistical analysis is performed by one-way ANOVA followed by Tukey's multiple comparison test. (G) At 2dpf, 2,3-BDM prevented the CVP cavernoma dramatically. 164 embryos in 2,3-BDM treated group and 177 in control group were used for Two-tailed paired t-test. P=0.0013.

protected against the formation of CCMs, suggesting that propranolol's therapeutic effect on CCM is through β1AR antagonism. Together with the observed significant reduction of CCM lesion volume upon metoprolol treatment, a selective β1AR antagonist, supports the therapeutic potential of β1AR antagonism in CCMs. β1 selective blockers offer the advantage of causing fewer mechanism-based side effects compared to propranolol, such as bronchospasm[22], and are already in clinical use (including agents like atenolol, metoprolol, nebivolol, and bisoprolol). Thus, further studies are warranted to evaluate the potential value of β1 selective antagonists in this disease.

Consistent with the *Adrb1KO* mice[14], the *adrb1*$^{-/-}$ zebrafish embryos displayed the decreased chronotropic response to a beta-adrenergic agonist, isoprenaline, and decreased basal heart rate compared to that of *adrb1*$^{+/+}$ brood mates; however, CVP morphology of *adrb1*$^{-/-}$ embryo was not perturbed (Supplementary Figure 3). *adrb1*$^{-/-}$ embryo also displayed a marked decrease of blood flow in CVP (Supplementary Video 2, and Supplementary Figure 1B), A similar reduction of heart rate and contractility by 2,3-BDM, a cardiac myosin ATPase inhibitor, prevented CVP dilation (Figure.4F). We previously found that arresting blood flow prevents aberrant intussusceptive angiogenesis and the resulted CVP cavernomas in *ccm2* CRISPR embryos[3]. Taken together, these data suggest that reduced blood flow secondary to reduced cardiac function accounts for the protective effect of loss of β1AR signaling on CVP lesions and on CCM. To further confirm that reduced blood flow underlies the role of β1AR antagonism in rescuing these vascular defects, chemicals such as cardiac glycosides or phosphodiesterase inhibitors could be employed to restore cardiac pumping function in *adrb1*$^{-/-}$ embryos. Importantly, although β1AR are highly expressed in cardiomyocytes and contribute to increased cardiac output[23], β1ARs are also expressed in other tissues[24, 25] including endothelial cells of vascular anomalies in patients[26]. Thus, future studies will be required to delineate the tissue-specific contributions of β1AR signaling to CCM pathogenesis.

Experimental procedures

Zebrafish lines and handling

Zebrafish were maintained and with approval of Institutional Animal Care and Use Committee of the University of California, San Diego. The following mutant and transgenic lines were maintained under standard conditions: *Tg(fli1:EGFP)*^{y1}[27], *Tg(klf2a:H2b-EGFP)*[28], and *Tg(kdrl:mcherry)*^{is5}[29]. *adrb1*$^{-/-}$ zebrafish was obtained by coninjection of Cas9 protein (EnGen Spy Cas9 NLS, M0646, NEB) with gRNA targeting *adrb1*. Genotyping of *adrb1*$^{-/-}$ was performed with forward primer (5'-AGAGCAGAGCGCGGATGGAA-3') and reverse primer (5'-GATCCATACATCCAGGCT-3').

Plasmids and morpholino

pCS2-nls-zCas9-nls (47929) and pT7-gRNA (46759) were bought from Addgene. The CRISPR RNA (crRNA) sequences used in this study are as follow: *ccm2*-1 5'-GGTGTTTCTGAAAGGGGAGA-3', *ccm2*-2 5'-GGAGAAGGGTAGGGATAAGA-3', *ccm2*-3 5'-GGGTAGGGATAAGAAGGCTC-3', *ccm2*-4 5'-GGACAGCTGACCTCAGTTCC-3', adrb1 5'-GACTCTAAACGCGCCACGG-3'. Target gRNA constructs were generated as described before[30]. Morpholino sequence used in this study are: *adrb1* (5'-ACGGTAGCCCGTCTCCCATGATTTG-3')[31], *ccm2* (5'-GAAGCTGAGTAATACCTTAA CTTCC-3')[32], *tnnt2a* (5'-CATGTTTGCTCTGATCTGACACGCA-3')[33], control (5'-CCTCTTACCTCAGTTACAATTTATA-3').

RNA synthesis

The pCS2-nls-zCas9-nls plasmid containing Cas9 mRNA was digested with NotI enzyme, followed by purification using a Macherey-Nagel column, serving as the template. The capped nls-zCas9-nls RNA was synthesized using the mMESSAGE mMACHINE SP6 Transcription Kit from ThermoFisher Scientific. The resulting RNA was purified through lithium chloride precipitation, as per the instructions provided in the kit. For gRNA synthesis, the gRNA constructs were linearized using BamHI enzyme and purified using a Macherey-Nagel column. The gRNA was synthesized via in vitro transcription using the MEGAshortscript T7 Transcription Kit from ThermoFisher Scientific. After synthesis, the gRNA was purified by alcohol precipitation, as instructed in the same kit. The concentration of the nls-zCas9-nls RNA and gRNA was measured using a NanoDrop 1000 Spectrophotometer from ThermoFisher Scientific, and their quality was confirmed through electrophoresis on a 1% (wt/vol) agarose gel.

Microinjection

All injections were performed at 1-cell stage with a volume of 0.5nl. The final injection concentrations are as follow: Cas9 protein (10μM), Cas9 mRNA (750ng/μl), gRNA 120ng/μl, *adrb1* MO (4ng/μl), *ccm2* MO (4ng/μl), *tnnt2a* MO (5.3 ng/μl), control MO (4ng/μl).

Chemical treatment

Propranolol (P0995, TCI) (12.5μM) and metoprolol (M1174, Spectrum) (50μM) were used to treat the zebrafish larva from Day 21. Beginning from Day 35, adjusted concentration of propranolol (25μM) or metoprolol (100μM) were used to treat the juvenile fish. The chemicals were dissolved in fish water, and fish water containing chemicals were refreshed daily. Egg water without above chemicals was refreshed daily for fish used as negative control. 2,3-butanedione monoxime (BDM) (6mM) was added to egg water of the *ccm2* CRISPR embryos at 25hpf, and CVP cavernoma was observed at 2dpf.

Airyscan imaging and fluorescence intensity analysis

To prepare the embryos for imaging, they were first anesthetized using egg water containing 0.016% tricaine (3-amino benzoic acid ethyl ester) from Sigma-Aldrich. Subsequently, the anesthetized embryos were embedded in 1% low melting point agarose obtained from Invitrogen (product number 16520050). The imaging process was carried out using a Zeiss 880 Airyscan confocal microscope, utilizing the standard Airyscan stack mode, with a Plan-Apochromat 20x/0.8 M27 objective. The scanning setup is as follow: Lasers (Green 488nm: 26.0%, Red 561nm: 15.0%), Master Gain (800), Digital Gain (1.00), Scaling X (0.415μm), Scaling Y (0.415μm), and Scaling Z (0.800μm). The intensity of nuclear EGFP (enhanced green fluorescent protein) was quantified using ImageJ software. The selected background area signal was measured by running Analyze>Measure. Then the background value was subtracted by running Process>Math>Subtract. “Freehand selections” button was used for outlining the endothelial nucleus stack by stack along Z-axis. By running Analyze>Measure, the information of “Area” and “IntDen (Integrated Density)” of the selected endothelial nucleus was obtained. The average EGFP intensity of a nucleus equals to the summation of “IntDen” divided by summation of “Area”.

Heartbeat and blood flow recording

Heartbeat and blood flow were recorded using Olympus MVX10. The embryos were treated with 0.004% tricaine which does not have effect on heartbeat[34, 35].

Zebrafish brain dissection, CUBIC treatment and lightsheet imaging

The dissection of zebrafish brains followed the methodology described in a previous study by Gupta et al.[36]. The CUBIC method was optimized based on the findings from a previous report by Susaki et al.[15]. The brains were fixed in 4% paraformaldehyde (PFA) with a pH of 7.5 for 24 hours and subsequently washed with PBS (phosphate-buffered saline) for an additional 24 hours. Following the PBS wash, the brains underwent CUBICR1 treatment at 37°C in a water bath for 42 hours. For imaging, the samples were placed in CUBICR2 as the imaging medium and imaged using a ZEISS Lightsheet Z.1 microscope. Scanning was carried out utilizing 5X dual illumination optics in combination with a 5X objective.

Statistical analysis

The statistical analysis was conducted using GraphPad Prism software. P-values were calculated using an unpaired two-tailed Student's t-test, unless otherwise specified. The bar graphs display the mean values along with their corresponding standard deviations (SD).

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Additional files

Supplementary Figure Legends

Supplementary Figures and Tables

Supplementary Video 1 adrb1-/-

Supplementary Video 1 wild type

Supplementary Video 2 adrb1-/-

Supplementary Video 2 wild type

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Reviewer #2 (Public review):

Summary:

Previously, the authors developed a zebrafish model for cerebral cavernous malformations (CCMs) via CRISPR/Cas9-based mosaic inactivation of the ccm2 gene. This model yields CCM-like lesions in the caudal venous plexus of 2 days post-fertilization embryos and classical CNS cavernomas in 8-week fish that depend, like the mouse model, on the upregulation of the KLF2 transcription factor. Remarkably, the morpholino-based knockdown of the gene encoding the Beta1 adrenergic receptor or B1AR (adrb1; a hemodynamic regulator) in fish and treatment with the anti-adrenergic S enantiomer of propranolol in both fish and mice reduce the frequency and size of CMM lesions.

In the present study, the authors aim to test the model that adrb1 is required for CCM lesion development using adrb1 mutant fish (rather than morpholino-mediated knockdown and pharmacological treatments with the anti-adrenergic S enantiomer of propranolol or a racemic mix of metoprolol (a selective B1AR antagonist).

Strengths:

The goal of the work is important, and the findings are potentially highly relevant to cardiovascular medicine.

Comments on latest version:

This reviewer is largely satisfied and congratulates the authors on their updated work. However, the comments regarding the caveats of morpholino use and lack of validation that the morphants phenocopy the mutants using the readouts that they employ still stand (for instance, the tnnt2a MO has been extensively validated for phenocopying lack of cardiac contractility, not for the phenotypes under study). Finally, while using the cytosolic red line to mask a nuclear green readout is suboptimal (not for FRET reasons), this is now a minor issue given that all comparisons are made using this method and the increase in sample size.

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Author response:

The following is the authors' response to the original reviews.

> ***Reviewer #1 (Public review):***
>
> *Summary:*
>
> *This work seeks to provide genetic evidence for a role for beta-adrenergic receptors that regulate heart rate and blood flow on cavernous malformation development using a*

> *zebrafish model, and to extend information regarding beta-adrenergic drug blockade in cavernous malformation development, with the idea that these drugs may be useful therapeutically.*
>
> *Strengths:*
>
> *The work shows that genetic loss of a specific beta-adrenergic receptor in zebrafish, adrb1, prevents embryonic venous malformations and CCM in adult zebrafish brains. Two drugs, propranolol and metoprolol, also blunt CCM in the adult fish brain. These findings are predicted to potentially impact the treatment of human CCM, and they increase understanding of the factors leading to CCM.*

Response 1: We are grateful for the reviewer's acknowledgment of this study's potential translational significance.

> *Weaknesses:*
>
> *There are minor weaknesses that detract slightly from enthusiasm, including poor annotation of the Figure panels and lack of a baseline control for the study of Klf2 expression (Figure 4).*

Response 2: We agree. Annotation of the Figure panels were added, and a baseline control for the study of *klf2a* expression (Figure 4) was added. Details were described in the response to "recommendations for the authors".

> ***Reviewer #2 (Public review):***
>
> *Summary:*
>
> *Previously, the authors developed a zebrafish model for cerebral cavernous malformations (CCMs) via CRISPR/Cas9-based mosaic inactivation of the ccm2 gene. This model yields CCM-like lesions in the caudal venous plexus of 2 days post-fertilization embryos and classical CNS cavernomas in 8-week fish that depend, like the mouse model, on the upregulation of the KLF2 transcription factor. Remarkably, the morpholino-based knockdown of the gene encoding the Beta1 adrenergic receptor or B1AR (adrb1; a hemodynamic regulator) in fish and treatment with the anti-adrenergic S enantiomer of propranolol in both fish and mice reduce the frequency and size of CMM lesions.*
>
> *In the present study, the authors aim to test the model that adrb1 is required for CCM lesion development using adrb1 mutant fish (rather than morpholino-mediated knockdown and pharmacological treatments with the anti-adrenergic S enantiomer of propranolol or a racemic mix of metoprolol (a selective B1AR antagonist).*
>
> *Strengths:*
>
> *The goal of the work is important, and the findings are potentially highly relevant to cardiovascular medicine.*

Response 3: We are grateful for the reviewer's acknowledgment of this study's scientific importance and clinical relevance.

> *Weaknesses:*
>
> *(1) The following figures do not report sample sizes, making it difficult to assess the validity of the findings: Figures 1B and D (the number of scored embryos is missing), Figures 2G and 3B (should report both the number of fish and lesions scored, with color-coding to label the lesions corresponding to individual fish in which they were found).*

Response 4: We agree. Sample sizes of Figures 1B and D were added in the figures and figure legends. Sample sizes of Figures 2G and 3B were added in their figure legend respectively. The lesion volume in Figures 2G and 3B is the total lesion volume in each brain.

> *(2) Figure 4 has a few caveats. First, the use of adrb1 morphants (rather than morphants) is at odds with the authors' goal of using genetic validation to test the involvement of adrb1 in CCM2-induced lesion development.*

Second, the authors should clarify if they have validated that the tnnt (tnnt2a) morpholino phenocopies tnnt2a mutants in the context in which they are using it (this reviewer found that the tnnt2a morpholino blocks the heartbeat just like the mutant, but induces additional phenotypes not observed in the mutants).

Response 5: We appreciate the reviewer's comments; however, generating *adrb1*$^{-/-}$ and *tnnt2a*$^{-/-}$ *klf2a* reporter fish, while also ensuring the presence of only one EGFP transgene allele for intensity measurement, would require prohibitively time-consuming breeding efforts.

The use of morpholinos for *tnnt2a* and *adrb1*, as well as their effects on the heart, have been well-documented in previous studies (Sehnert AJ et al., *Nat Genet.* 2002;31:106-10; Steele SL et al., *J Exp Biol.* 2011;214:1445-57).

> *Third, the data in Figure 4E is from just two embryos per treatment, a tiny sample size. Furthermore, judging from the number of points in the graph, only a few endothelial PCV cells appear to have been sampled per embryo. Also, judging from the photos and white arrowheads and arrows (Figure 4A-D), only the cells at the ventral side of the vessel were scored (if so, the rationale behind this choice requires clarification).*

Response 6: We have increased the sample size, as described in the Figure 4 legend. Regarding the scoring of endothelial nuclei, we focused on the ventral side of the vessel because nuclei on the dorsal side often reside at branching points of the venous plexus. This positional variance could influence *klf2a* expression levels; thus, we focused on the ventral surface to limit this potential confounding variable.

> *Fourth, it is unclear whether and how the Tg(kdrl:mcherry)is5 endothelial reporter was used to mask the signals from the klf2a reporter. The reviewer knows by experience that accuracy suffers if a cytosolic or cell membrane signal is used to mask a nuclear green signal.*

Response 7: We agree that it is theoretically possible for Förster resonance energy transfer (FRET) to occur, as the emission spectrum of EGFP (495-550 nm in our filter setup) overlaps with the absorption spectrum of mCherry. However, several factors reduce the likelihood of FRET in our experimental setup:

(1) Without a nuclear localization signal, the majority of mCherry is localized in the cytoplasm, although small amounts may passively diffuse into the nucleus.

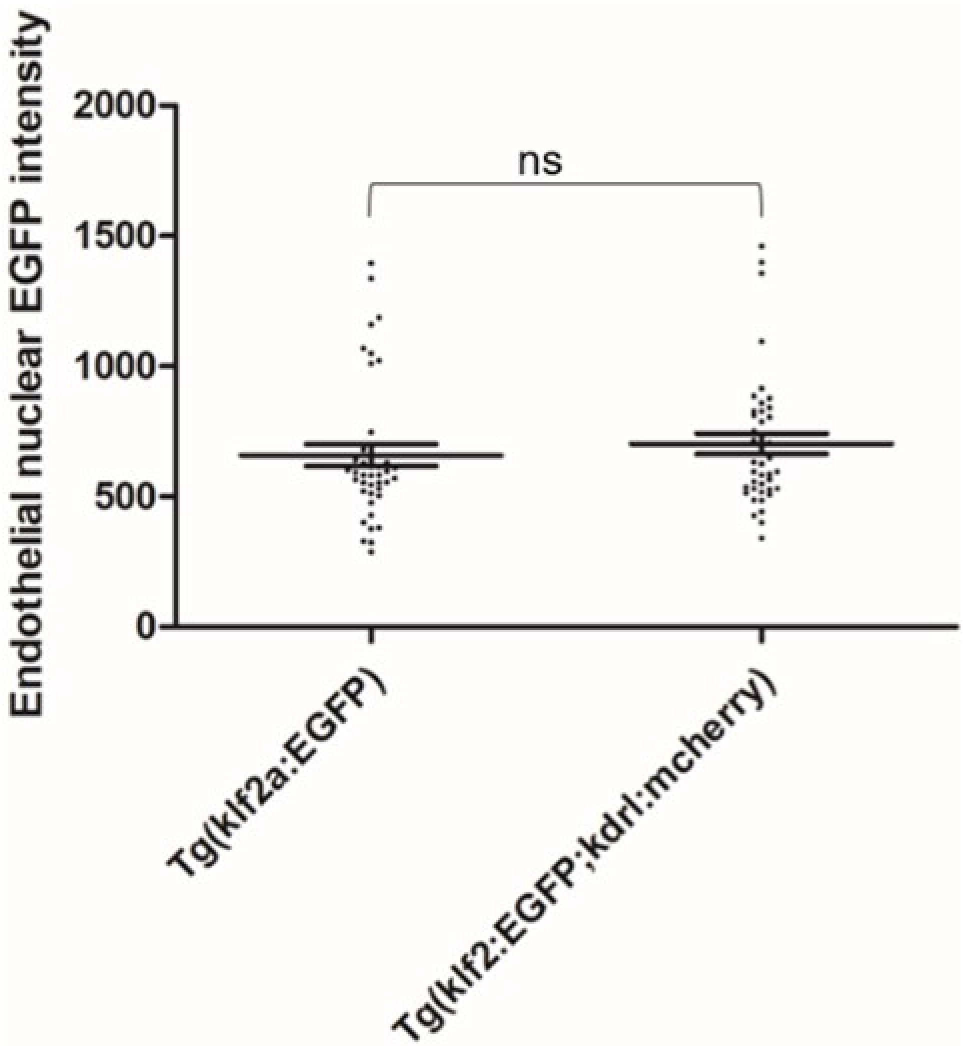
(2) EGFP, on the other hand, is predominantly localized in the nucleus due to the presence of a nuclear localization signal.

(3) FRET requires two fluorophores to be within a proximity of 8-10 nanometers or less for efficient energy transfer. The nuclear envelope, with a typical thickness of 30-50 nanometers, separates nuclear EGFP from cytoplasmic mCherry and FRET efficiency is inversely proportional to the sixth power of the distance between donor and acceptor. Thus, the theoretical likelihood of significant energy transfer under these conditions is low.

To empirically examine potential FRET between nuclear EGFP and mcherry in our experiment setup, we scanned and scored the Tg(klf2a:H2b-EGFP; kdrl:mcherry) double transgenic embryos and Tg(klf2a:H2b-EGFP) embryos for EGFP intensity. The result is attached here:

Author response image 1.

42 endothelial nuclei from 7 embryos were scored as described in the Experimental Procedures of the manuscript. Two tailed t test were performed. P=0.4529



> *Finally, the text and legend related to Figure 4 could be more explicit. What do the authors mean by a mosaic pattern of endothelial nuclear EGFP intensity, and how is that observation reflected in graph 4E? When I look at the graph, I understand that klf2a is decreased in C-D compared to A-B. Are some controls missing? Suppose the point is to show mosaicism of Klf2a levels upon ccm2 CRISPR. Don't you need embryos without ccm2 CRISPR to show that Klf2a levels in those backgrounds have average levels that vary within a defined range and that in the presence of ccm2 mosaicism, some cells have values significantly outside that range? Also, in 4A-D, what are the white arrowheads and arrows? The legend does not mention them.*

Response 8: We have revised our description of Figure 4 to better convey that mosaic expression of KLF2a is evidenced by the wide variability of *klf2a* reporter intensity in endothelial cells in *ccm2* CRISPR embryos. A baseline control for the study of *klf2a* expression was added to Figure 4. The arrowheads and arrows in Figure 4A-D are explained in Figure 4 legends.

> *Given the practical relevance of the findings to cardiovascular medicine, increasing the strength of the evidence would greatly enhance the value of this work.*
>
> ***Recommendations for the authors:***
>
> ***Reviewing Editor:***
>
> *Concerns about the labeling of figures and sample sizes should both be addressed, as detailed in the reviews, as this will be important to ensure the robustness of the claims.*
>
> ***Reviewer #1 (Recommendations for the authors):***
>
> *Overall a strong research advance that provides rigorous genetic analysis and further drug testing in the zebrafish CCM model. There are some minor issues that, if addressed, would strengthen the work.*
>
> *Minor issues:*
>
> *(1) Figures in general are very poorly annotated and labeled. None of the images in Figures 1-3 show the reporter used to visualize vessels/CM, and the scale bars are not sized in the Figures or legends. Figure 1B is an experiment where the effects of a drug that increases heart rate are evaluated in mutants and controls, but the drug is not mentioned in the figure panel. Figure 1D shows the percentage of embryos with CVP dilation, but the graph and accompanying description does not define whether the percent is relative to the total embryos from the intercross or the percent of that category having the CVP dilation.*

Response 9: Changes were made in Figures and Figure legends. The transgenic reporter line Tg(fli1:EGFP) was annotated in Figures 1-3. Scale bars were sized in the Figures and Figure legends. The chemical used for Figure 1B was annotated in the Figure. The percentage of CVP dilation in the graph was explained in the Figure legend.

> *(2) Figure 4 does not include baseline data in unmanipulated embryos scored at the same time to show the increase in Klf2 expression with mosaic ccm2 deletion. This is important as the result in E is interpreted as a lack of change in the increase.*

Response 10: A baseline control for the study of *klf2a* expression in Figure 4 was added.

> ***Reviewer #2 (Recommendations for the authors):***
>
> *SUGGESTIONS FOR EXPERIMENTS, DATA, OR ANALYSES*
>
> *(1) For maximum rigor, in the Figure 4 experiment, use adrb1 mutants and tnnt2a (silent heart) mutants (or verify that the adrb1 and tnnt2a morpholinos faithfully copy the phenotype of interest). See: Guidelines for morpholino use in zebrafish (PMID: 29049395; PMCID: PMC5648102).*

Response 11: See Response 5.

> *(2) Increase sample sizes if appropriate.*

Response 12: In the revised version of the manuscript, we have increased the sample size, as described in the Figure 4 legend.

> *(3) The imaging and fluorescence intensity analysis methods require more detail for reproducibility's sake. Please provide this information. See as a guideline: Guillermo MarquésThomas PengoMark A Sanders (2020) Science Forum: Imaging methods are vastly underreported in biomedical research eLife 9:e55133.*

Response 13: We added detailed procedures for the “Airyscan imaging and fluorescence intensity analysis” in the “Experimental Procedures”.

> *(4) I suggest further clarifying how inhibition of B1AR prevents cavernoma formation. Given that lesion formation is suppressed in adrb1 mutants (which have slow blood flow) and 2,3-BDM treatment (which also slows blood flow) has a similar effect, the beneficial effects of propranolol and metoprolol might be due to the slowing of blood flow via B1AR targeting rather than reflecting that B1AR is a critical component of the genetic circuit for cavernoma formation. Indeed, in prior work by the same first author and collaborators (Elife 2021 May 20:10:e62155), the investigators observed reduced cavernoma formation in embryos devoid of cardiac contractility and thus lacking blood flow (tnnt2a morphants). Such a scenario does not take away the value of a pharmacological treatment. Still, it implies a different mechanism and allows potentially many other drugs with similar effects on blood flow to be effective.*
>
> *Discussing how B1AR activity is regulated and outlining future experiments would be helpful. Suggestions for the latter include testing the effect of normalizing blood flow in adrb1 mutants with a drug or providing exogenous B1AR in the myocardium or the endothelium to test the model further.*

Response 14: We are grateful for the reviewer’s suggestions and added the statement for future experiments.

> *MINOR CORRECTIONS TO TEXT AND FIGURES*
>
> *(1) Figure 4E: Label the four genotypes explicitly, rather than A-D for the reader's ease.*
>
> *(2) Legend of Figure 4: "(F) EGFP intensity...". It should be (E).*
>
> *CITATIONS TO CORRECT*
>
> *(1) The citation for the Tg(kdrl:mcherry)is5 transgene needs to be corrected (reference 29 is from the Stainier lab). However, the "is" designation is for the Essner lab (https://zfin.org/action/feature/view/ZDB-ALT-110127-25)*

Response 15: Corrections were made as instructed.

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